MITTAL SCHOOL OF BUSINESS

Course Code:- INTM590	**Course Title:- BIG DATA ANALYTICS**
Course Instructor:- Ms. Naman Saini	
Academic Task No.:- 1	**Academic Task Title:** Project
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Student's Roll no:- RQ5E07A13	**Student's Reg. no:-** 12501430
Evaluation Parameters:	

Declaration:

I declare that this Assignment is my individual work. I have not copied it from any other student's work or from any other source except where due acknowledgement is made explicitly in the text, nor has any part been written for me by any other person. **Student's Signature: Soni Kumari**

Evaluator's comments (For Instructor's use only) Evaluator's Signature and Date:

General Observations	Suggestions for Improvement	Best part of assignment

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty member **Ms. Naman Saini**, for their valuable guidance, continuous support, and encouragement throughout the completion of this project on Big Data Analytics. Their expert insights and practical suggestions helped me understand the real-world implementation of advanced tools such as Sqoop.

I am deeply thankful to Lovely Professional University and the Mittal School of Business for providing the academic environment, technical resources, and practical exposure necessary to successfully complete this assessment. The university's structured curriculum and hands-on learning approach significantly contributed to enhancing my understanding of big data technologies.

I would also like to acknowledge the availability of system resources, software tools, and datasets that made it possible to perform various practical tasks, including data storage in HDFS, data migration through Sqoop.

Special thanks to my peers and mentors who supported me with constructive discussions and motivation whenever required.

Lastly, I am grateful for the opportunity to work on this project, as it has strengthened my technical knowledge, improved my practical skills, and provided me with valuable experience in handling big data tools and technologies.

This project has been a meaningful learning experience, and I am proud to have completed it through my own sincere efforts.

CERTIFICATE

This is to certify that **Soni Kumari**, a student of **Lovely Professional University**, enrolled in **Big Data Analytics (Course Code: INTM590)** under the **Mittal School of Business, Lovely Faculty of Business and Arts**, has successfully completed the project titled **"Implementation of Big Data Tools using Sqoop"** during the academic session assigned.

The work submitted in this report is an original record of the practical knowledge, technical understanding, and individual effort demonstrated by the student under the guidance of the faculty member. This project includes practical implementation of data import/export operations using Sqoop.

To the best of my knowledge, this work has been carried out sincerely by the student and fulfils the academic requirements prescribed by Lovely Professional University for the course. The student has shown dedication, analytical ability, and practical understanding while completing this assignment.

I appreciate the consistent effort and commitment shown throughout the completion of this project and wish the student success in future academic and professional endeavours.

Signature of Supervising Teacher

(Mr. Robin Tyagi)

"Data Transfer between RDBMS and Hadoop using Apache Sqoop"

Introduction

In today's digital world, data is growing at an extremely fast pace. Organizations generate large amounts of structured and unstructured data every day through transactions, social media, sensors, and various online platforms. Traditional database systems like MySQL are efficient for handling structured data but are not capable of managing very large datasets effectively. This limitation led to the development of Big Data technologies such as Hadoop.

Hadoop is an open-source framework that allows distributed storage and processing of large datasets across clusters of computers. It uses HDFS (Hadoop Distributed File System) to store data and MapReduce for processing. However, organizations still rely on relational database systems (RDBMS) like MySQL, Oracle, and SQL Server for structured data storage. Therefore, there is a need for a tool that can efficiently transfer data between RDBMS and Hadoop.

Apache Sqoop is designed to solve this problem. Sqoop is a command-line tool that allows users to import data from relational databases into Hadoop and export data back from Hadoop to databases. It automates the data transfer process and ensures high performance through parallel processing.

This project focuses on understanding the working of Sqoop and implementing various Sqoop commands to transfer data from MySQL to Hadoop.

Objectives of the Project

The main objectives of this project are:

- To understand the concept of Big Data and Hadoop ecosystem
- To study the role of Apache Sqoop in data integration
- To establish a connection between MySQL and Hadoop
- To execute different Sqoop commands for importing data
- To learn how to filter and select specific data during import
- To analyze the efficiency of Sqoop in handling large datasets

Tools and Technologies Used

1 Hadoop (HDFS)

Hadoop is used for storing large datasets in a distributed manner. HDFS divides data into blocks and stores them across multiple machines.

2 Apache Sqoop

Sqoop is used for transferring structured data between MySQL and Hadoop.

3 MySQL (RDBMS)

MySQL is used as the source database where structured data is stored in tables.

4 Linux/Cloudera Environment

Commands are executed in a Linux-based environment, typically Cloudera.

Working of Apache Sqoop

Apache Sqoop works by connecting to relational databases using JDBC (Java Database Connectivity). It retrieves metadata of tables and generates MapReduce jobs to transfer data efficiently.

When a Sqoop import command is executed:

1. Sqoop connects to the database
2. It fetches table structure
3. It creates MapReduce jobs
4. Data is divided into chunks
5. Each mapper transfers a portion of data
6. Data is stored in HDFS

This parallel processing makes Sqoop fast and efficient.

Sqoop Commands with Explanation

1 List Databases in MySQL

Command:

sqoop list-databases --connect jdbc:mysql://localhost/ --username root --password cloudera

Explanation:
This command displays all the databases available in MySQL. It is the first step to check whether the connection between Sqoop and MySQL is successful. If the connection fails, an error will be displayed.

```
[cloudera@quickstart ~]$ sqoop list-databases --connect jdbc:mysql://localhost/
--username root --password cloudera
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
26/04/22 05:02:48 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6-cdh5.13.0
26/04/22 05:02:48 WARN tool.BaseSqoopTool: Setting your password on the command-
line is insecure. Consider using -P instead.
26/04/22 05:02:50 INFO manager.MySQLManager: Preparing to use a MySQL streaming
resultset.
information_schema
cloth
cm
firehose
hue
metastore
mysql
nav
navms
oozie
retail_db
rman
sentry
```

2 List Tables in a Database

Command:

sqoop list-tables --connect jdbc:mysql://localhost/retail_db --username root --password cloudera

Explanation:
This command lists all tables present in a specific database. It helps users identify which tables can be imported into Hadoop.

```
[cloudera@quickstart ~]$ sqoop list-tables --connect jdbc:mysql://localhost/reta
il_db --username root --password cloudera
Warning: /usr/lib/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
26/04/22 05:17:45 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6-cdh5.13.0
26/04/22 05:17:46 WARN tool.BaseSqoopTool: Setting your password on the command-
line is insecure. Consider using -P instead.
26/04/22 05:17:48 INFO manager.MySQLManager: Preparing to use a MySQL streaming
resultset.
categories
customers
departments
order_items
orders
products
```

3 Import a Table from MySQL to HDFS

Command:

sqoop import --connect jdbc:mysql://localhost/retail_db --username root --password cloudera --table products --target-dir /user/cloudera/hbase -m1

Explanation:
This command imports an entire table from MySQL into HDFS.

- --connect → database connection string
- --table → name of the table
- --target-dir → destination in HDFS
- -m1 → number of mappers (parallel tasks)

```
        File System Counters
                FILE: Number of bytes read=0
                FILE: Number of bytes written=171387
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=87
                HDFS: Number of bytes written=173993
                HDFS: Number of read operations=4
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Other local map tasks=1
                Total time spent by all maps in occupied slots (ms)=37333
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=37333
                Total vcore-milliseconds taken by all map tasks=37333
                Total megabyte-milliseconds taken by all map tasks=38228992
        Map-Reduce Framework
                Map input records=1345
                Map output records=1345
                Input split bytes=87
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=420
                CPU time spent (ms)=4810
                Physical memory (bytes) snapshot=136753152
                Virtual memory (bytes) snapshot=1510199296
                Total committed heap usage (bytes)=60882944
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=173993
26/04/22 06:01:16 INFO mapreduce.ImportJobBase: Transferred 169.915 KB in 139.44
73 seconds (1.2185 KB/sec)
26/04/22 06:01:16 INFO mapreduce.ImportJobBase: Retrieved 1345 records.
[cloudera@quickstart ~]$
```

4 Import Specific Data using WHERE Clause

Command:

sqoop import --connect jdbc:mysql://localhost/rr --username root --password cloudera --table rtable --where 'id=1' --target-dir /user/cloudera/rob2 -m1

Explanation:
This command imports only specific rows based on a condition. It is useful when only a subset of data is required.

5 Import All Tables

Command:

sqoop import-all-tables --connect jdbc:mysql://localhost/rr --username root --password cloudera -m 1

Explanation:
This command imports all tables from a database into Hadoop automatically. It saves time when dealing with multiple tables.

6 Import Specific Columns

Command:

sqoop import --connect jdbc:mysql://localhost/rr --username root --password cloudera --table rtable --columns 'id' --target-dir /user/cloudera/tttt -m1

Explanation:
This command imports only selected columns from a table instead of the entire dataset.

Step-by-Step Procedure

1. Start Hadoop services
2. Open terminal in Linux/Cloudera
3. Ensure MySQL is running
4. Create database and tables in MySQL
5. Insert sample data into tables
6. Use Sqoop commands to connect MySQL
7. Execute import commands
8. Verify data in HDFS using:

hdfs dfs -ls /user/cloudera/

Advantages of Sqoop

- Provides fast data transfer using parallel processing
- Easy to use with simple commands
- Reduces manual data migration efforts
- Supports incremental data transfer
- Compatible with multiple databases

Limitations of Sqoop

- Only works with structured data
- Requires proper configuration of Hadoop and database
- Not suitable for real-time streaming data
- Performance depends on system configuration

Applications of Sqoop

- Data warehousing
- Big Data analytics
- Data migration projects
- Integration of enterprise systems

Conclusion

In conclusion, Apache Sqoop plays a vital role in the Hadoop ecosystem by enabling efficient data transfer between relational databases and distributed storage systems. In this project, we explored various Sqoop commands such as listing databases, importing tables, filtering data, and selecting specific columns. These operations demonstrate how Sqoop simplifies the process of data migration.

The ability to transfer large datasets quickly and efficiently makes Sqoop an essential tool for Big Data professionals. However, it is important to understand its limitations and use it appropriately based on the type of data and requirements. Overall, this project highlights the importance of Sqoop in bridging the gap between traditional database systems and modern Big Data technologies.